

# Introduction to R programming language

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**Advanced Data Analysis (ADA) Unit**

12<sup>th</sup> January 2026

# Course Outline

- **Day 1**

- R Basics – Operations, Packages & Classes
- Loops & Conditions
- Data Manipulation & Plotting
- Functions in R – From use to development

- **Day 2**

- Best Practices (Documentation, Reproducibility, ...)
- AI & Coding
- tidyverse Universe



# Programming & Research

- **Transparency**
- **Reproducibility**
- **Up-to-date methodology**
- **Big Data Era**



# Programming & Researcher

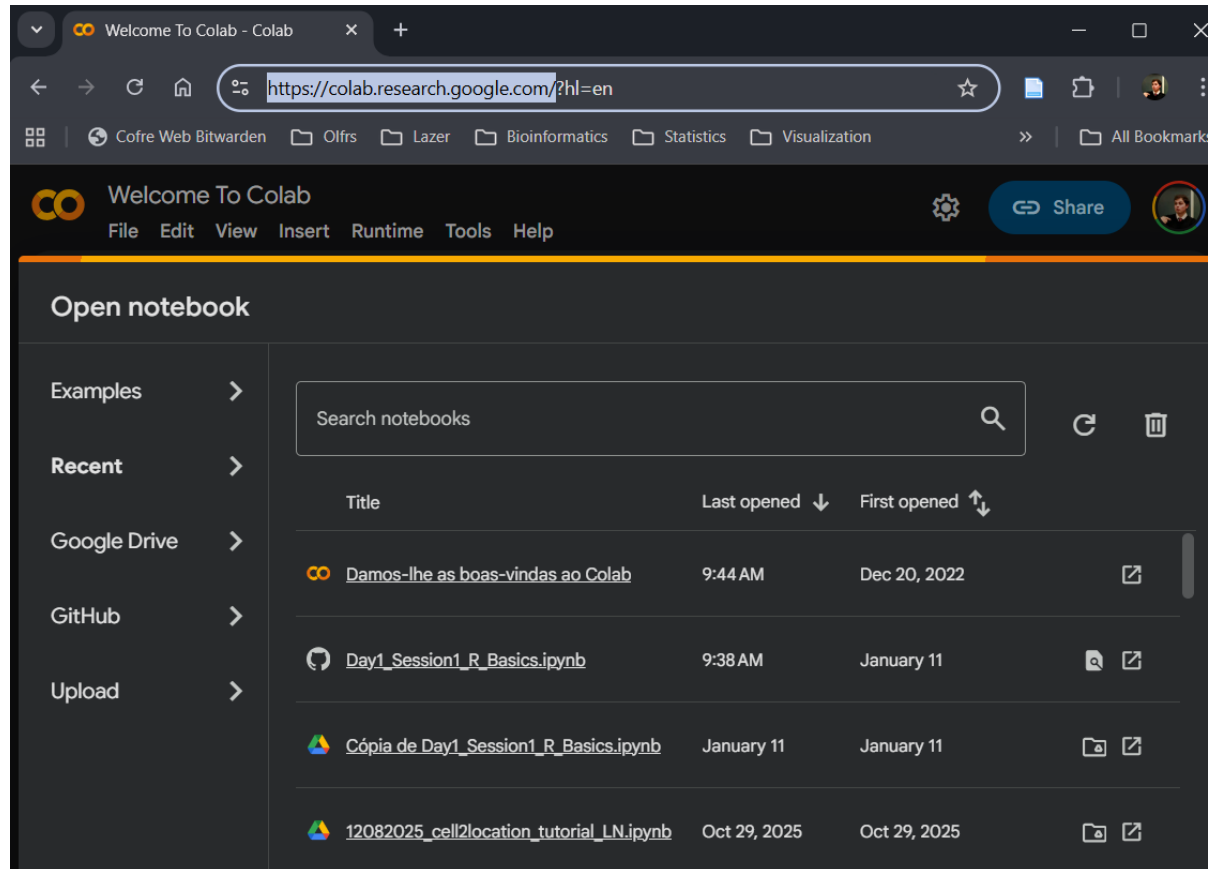
- **Documentation**
  - Citations!
  - Job market!
- **Capability & Flexibility**
- **Understanding Data Analysis**



# Course Material

[https://github.com/andreboilerbarros/Intro\\_R](https://github.com/andreboilerbarros/Intro_R)

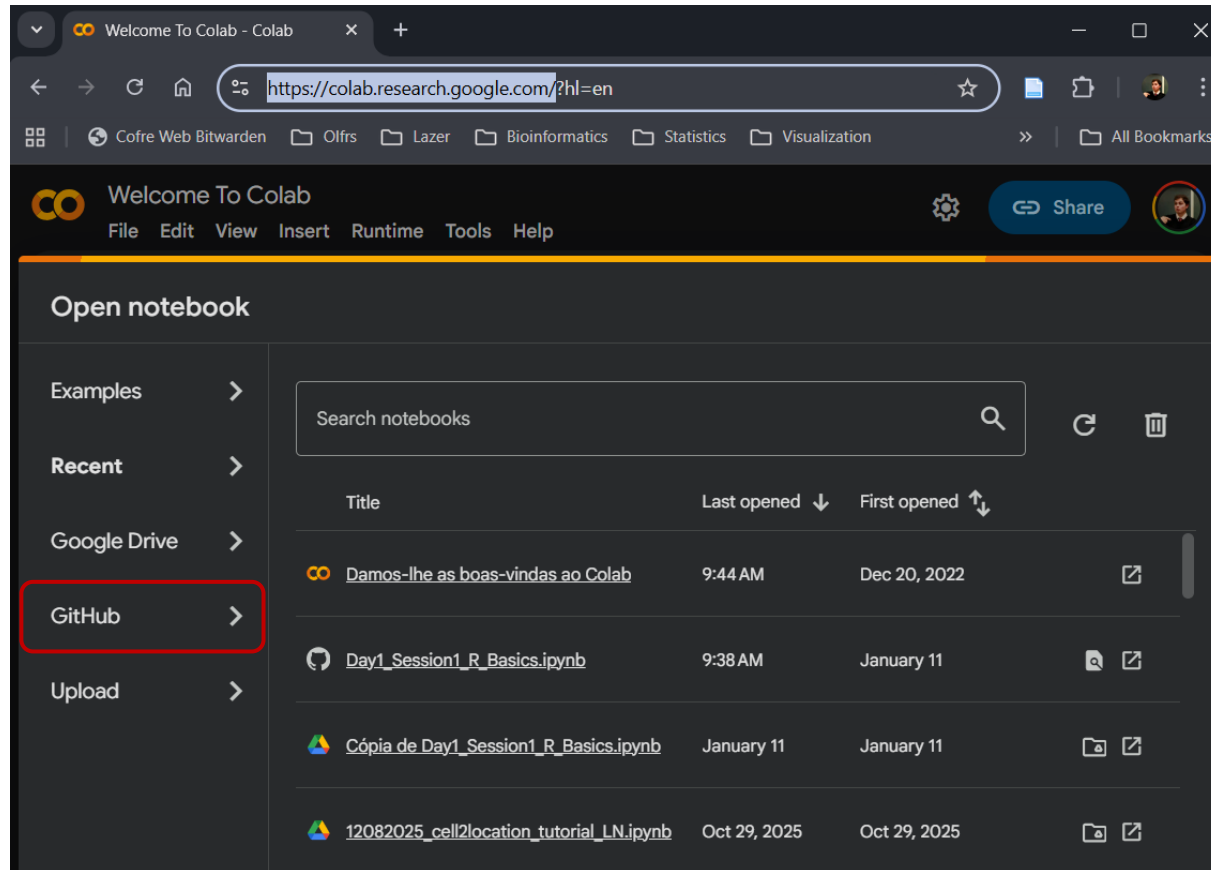
## 1. Google Colab – colab.research.google.com



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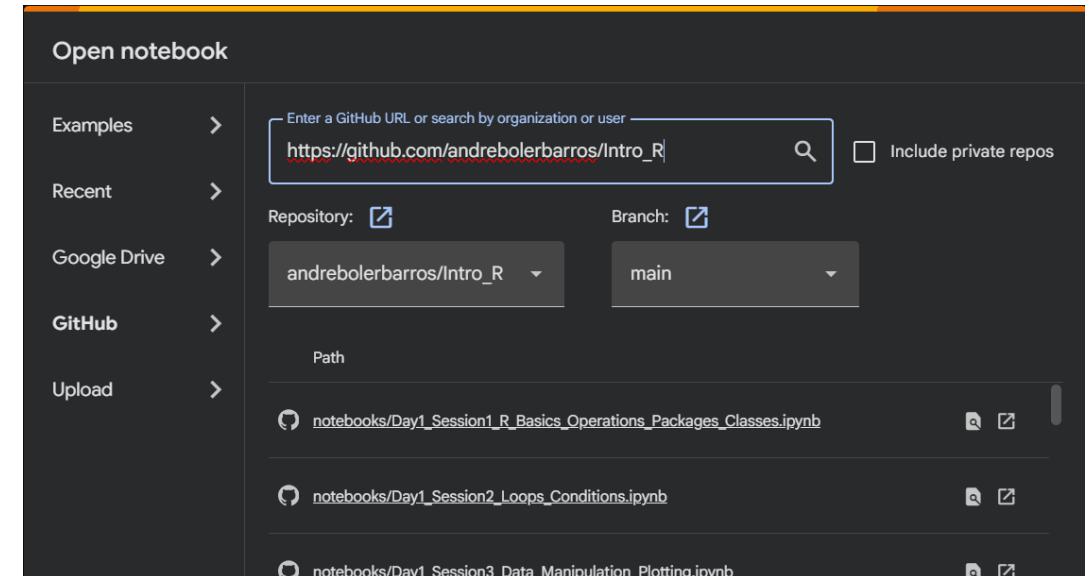
## 1. Google Colab – colab.research.google.com



Open notebook

Search notebooks

| Title                                    | Last opened ↓ | First opened ↑ |  |
|------------------------------------------|---------------|----------------|--|
| Damos-lhe as boas-vindas ao Colab        | 9:44 AM       | Dec 20, 2022   |  |
| Day1_Session1_R_Basics.ipynb             | 9:38 AM       | January 11     |  |
| Cópia de Day1_Session1_R_Basics.ipynb    | January 11    | January 11     |  |
| 12082025_cell2location_tutorial_LN.ipynb | Oct 29, 2025  | Oct 29, 2025   |  |



Open notebook

Enter a GitHub URL or search by organization or user

[https://github.com/andrebolbarros/Intro\\_R](https://github.com/andrebolbarros/Intro_R)

☐ Include private repos

Repository: Branch:

andrebolbarros/Intro\_R main

Path

notebooks/Day1\_Session1\_R\_Basics\_Operations\_Packages\_Classes.ipynb

notebooks/Day1\_Session2\_Loops\_Conditions.ipynb

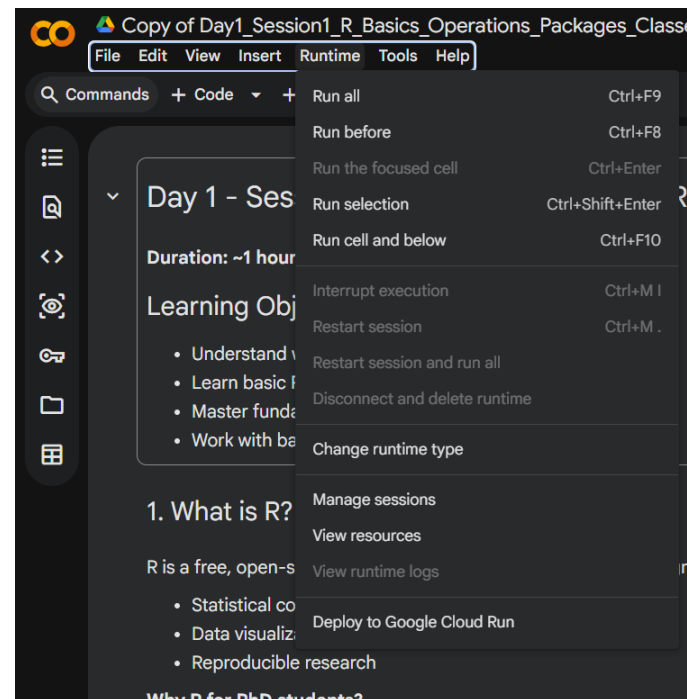
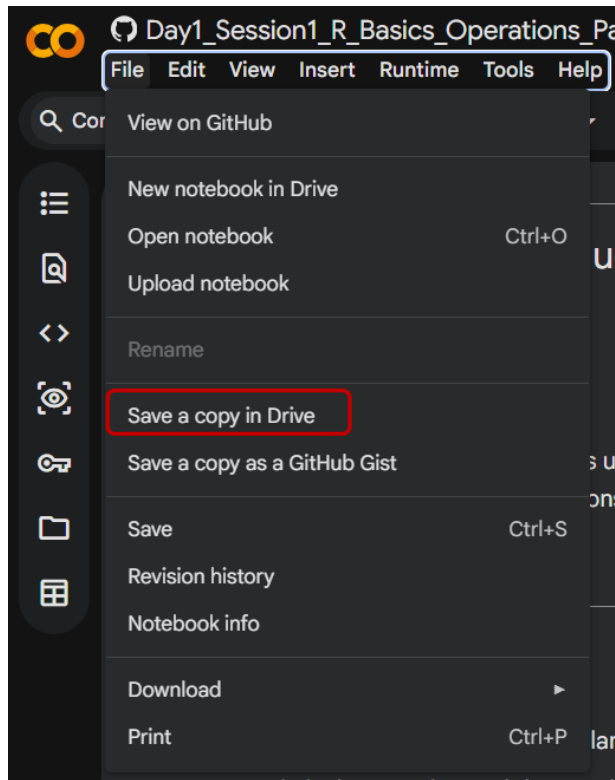
notebooks/Day1\_Session3\_Data\_Manipulation\_Plotting.ipynb



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## 2. Save your own copy



[https://github.com/andrebolbarros/Intro\\_R](https://github.com/andrebolbarros/Intro_R)

### 3. Change kernel (language)

